

# Exercises, Lecture #2

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## I Thresholds of singularities

1. Suppose that  $R$  is a Noetherian regular ring of characteristic  $p > 0$ . Prove that  $\text{fpt}(f) \leq 1$ .
2. Compute the  $F$ -pure threshold ( $\text{fpt}$ ) of  $xy(x - y) \in \mathbb{F}_2[x, y]$ .
3. Suppose that  $(R, \mathfrak{m})$  is a regular local ring essentially finite type over a field of characteristic zero and  $f \in \mathfrak{m}$ . Prove that  $\text{lct}(f) \leq 1$ .

4. Yesterday, you computed a log resolution  $Y \rightarrow \operatorname{Spec} k[x, y]$  of  $(k[x, y], y^2 - x^3)$ . Compute the sheaf  $\omega_Y$  for that map by using the following fact.

If you blowup a closed point on a 2-dimensional nonsingular variety (or scheme)  $X$  to obtain  $f : Y \rightarrow X$  with exceptional curve  $E$ , then  $\omega_Y = \mathcal{O}(E) \otimes f^* \omega_X$ .

5. Using the Grothendieck duality arguments from yesterday's exercises, plus trace arguments, one can prove the following in characteristic zero for  $f \in R$  (again,  $R$  is regular)

$$\operatorname{lct}(f) = \sup\{t > 0 \mid \Gamma(Y, \omega_Y \otimes \mathcal{O}_Y(t \operatorname{div}_Y(f))) = \omega_X\}.$$

Use this to compute the lct of  $y^2 - x^3 \in k[x, y]$ .

## II Macaulay2 exercises

Remember, you can run Macaulay2 online using the two links below if you don't have it installed on your computer.

- Macaulay2 at Georgia Tech
- Macaulay2 at University of Melbourne

### II.1 $F$ -pure thresholds in characteristic $p > 0$

Run the following commands.

```
needsPackage "TestIdeals"  
needsPackage "FrobeniusThresholds"
```

You can make a ring by the following command.

1. Make a ring and pick an element in it by running the following commands.

```
p = 5  
R = ZZ/p[x,y]  
f = x*y*(x-y)  
fpt(f)
```

Vary the  $p$ , make a conjecture about the value of  $\text{fpt}(f)$  when  $p \equiv 2 \pmod{3}$ .

2. Do the same for  $y^2 - x^3$ .

### II.2 Log canonical thresholds in characteristic zero

```
needsPackage "BernsteinSato"  
R = QQ[x,y,z]  
f = x^3+y^4+z^5
```

You can then run the following to compute the log canonical threshold ( $\text{lct}$ ).

```
lct(ideal(f))
```

Again, these computations are *not* done by computing resolutions of singularities. These are done via  $D$ -module techniques.

### II.3 Exercises

3. Do some experiments to make a prediction of what the  $\text{lct}$  of

$$x_1^{a_1} + \cdots + x_n^{a_n}$$

is. (At least if you don't already know the answer).

4. Compare the speed of computing  $\text{lct}$  in characteristic 0 with computing  $\text{fpt}$  in characteristic  $p > 0$ .